

SECTION 3.2 — Non-Functional Requirements

NFR-1: Performance

The system shall return the complete analysis (match score, gap list, and tailored content) within a reasonable processing time, targeted at under 30 seconds per request under normal conditions, accounting for the sequential execution of the four-agent pipeline.

NFR-2: Security

The system shall store user resumes and personal data securely in the database. Resume files and personal information shall not be exposed through any public or unauthenticated endpoint. Basic authentication shall be required to access a user's own resume and application data.

NFR-3: Quality/Reliability

The system shall produce consistent output for the same input; that is, given the same job description and resume, the match score and gap analysis should not vary significantly between runs. Each agent in the pipeline shall handle malformed or unexpected input (e.g., empty job description) without crashing the system.

NFR-4: Usability

The system shall provide a simple, easy-to-understand interface consisting of an input screen, a results screen, and a tracker dashboard, requiring no technical knowledge from the user. All generated content shall be presented in an editable format so users can make corrections without needing external tools.

NFR-5: Maintainability

The system's backend code shall be organized in a modular way, with each AI agent implemented as a separate, independently testable component, to allow future developers to modify or extend individual agents without affecting the rest of the pipeline.

NFR-6: Availability

The system shall remain accessible during scheduled demonstration and testing periods. Since the system depends on an external LLM provider, the system shall handle API failures gracefully by displaying an appropriate error message rather than crashing.